



VMware Cloud Foundation – STIG Hardening Overview

License Server
9.1.0.0

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Revision History

Date	Description of Change
May 2026	Initial Release

Overview

Broadcom is a trusted partner in highly secure, mission critical systems around the world, including the United States Department of Defense (DoD). In the DoD, all IT systems must adhere to the rigorous Risk Management Framework (RMF) as defined in DoDI 8510.01. A critical component of RMF is the mandatory implementation of Security Technical Implementation Guides (STIGs) and Security Requirements Guidelines (SRGs) as maintained by the Defense Information Systems Agency (DISA).

To support our customers, VCF License Server is evaluated against the following standards:

- VMware Cloud Foundation Photon OS 5.0 STIG Readiness Guide, Version 3 Release 3

This report will document the product's compliance with this guidance, including any deviations.

Applicability

The contents of this document are applicable to the following product versions:

- VMware Cloud Foundation 9.1.0.0 Build 25346031 License Server

Disclaimer

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VCF License Server Compliance

Licenses for VCF infrastructure and management components are stored on a dedicated license server appliance. The license server is a protected repository within the environment to store licensing data.

The license server is automatically deployed during new installations or upgrades. The lifecycle management of the license server is performed by VCF Operations.

The license server has been hardened by default and is not user-serviceable; updates to hardening are released as part of the product update and upgrade processes.

Photon OS 5.0 Compliance: Overall

The VCF License Server is built on Photon OS 5.0. The results below are based on auditing the appliance against the latest Photon OS 5.0 STIG Readiness Guide release.

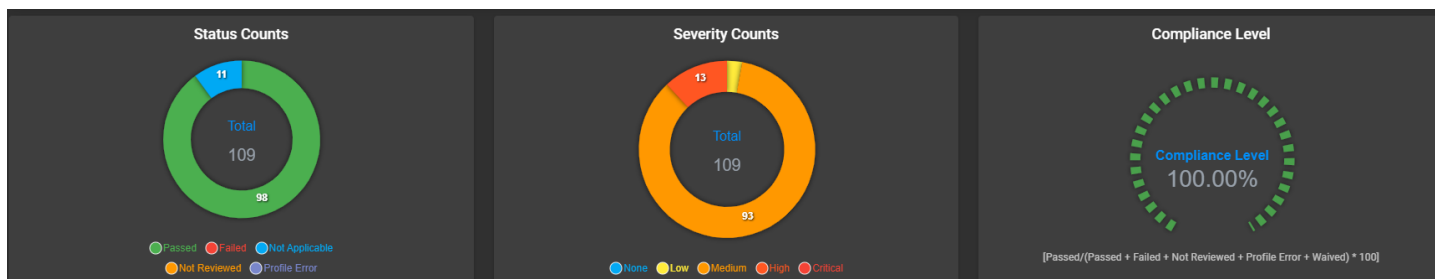


Figure 1: Overall Compliance for Photon OS 5.0

A full list of controls and their statuses is available in the Appendix section of this document.

Photon OS 5.0 Compliance: Not Applicable or Waived

The items listed in the following table are controls that are either Not Applicable to the appliance, or have been waived for reasons listed in the Justification column.

Control ID	Title	Justification
PHTN-50-000005	The Photon operating system must display the Standard Mandatory DOD Notice and Consent Banner before granting local or remote access to the system.	SSH is not present on the appliance.
PHTN-50-000047	The Photon operating system must disable unnecessary kernel modules.	Squashfs is required on the appliance.
PHTN-50-000069	The Photon operating system must terminate idle Secure Shell (SSH) sessions after 15 minutes.	SSH is not present on the appliance.
PHTN-50-000192	The Photon operating system must be configured to use the pam_faillock.so module.	SSH is not present on the appliance.
PHTN-50-000203	The Photon operating system must terminate idle Secure Shell (SSH) sessions.	SSH is not present on the appliance.
PHTN-50-000218	The Photon operating system must configure Secure Shell (SSH) to ignore user-specific known_host files.	SSH is not present on the appliance.
PHTN-50-000221	The Photon operating system must configure Secure Shell (SSH) to restrict LoginGraceTime.	SSH is not present on the appliance.

Control ID	Title	Justification
PHTN-50-000231	The Photon operating system must not perform IPv4 packet forwarding.	The appliance acts as a container host and IPv4 packet forwarding is required.
PHTN-50-000245	The Photon operating system must mount /tmp securely.	No interactive access is allowed to the appliance.
PHTN-50-000266	The Photon operating system must generate audit records for all account creations, modifications, disabling, and termination events that affect /etc/security/opasswd.	No interactive access is allowed to the appliance.
PHTN-50-000267	The Photon operating system must be configured to use the pam_deny.so module.	SSH is not present on the appliance.

Frequently Asked Questions

Can customers make changes to the vSphere Supervisor?

No. Due to the immutable nature of vSphere Supervisor deployments any changes would not be persistent. Modification to images and components is not supported. For product improvements please contact your VMware Account Team.

Where can I find the Kubernetes STIG?

The Kubernetes STIG may be found at:
<https://public.cyber.mil/stigs/>

Where can I find the Photon OS 5.0 STIG Readiness Guide?

The Photon OS 5.0 STIG Readiness guide may be found at:
<https://github.com/vmware/dod-compliance-and-automation/tree/master/photon/5.0/docs>

Is there a compliance report for vSphere Kubernetes Release (VKr) Clusters?

A report will be available in a separate document and those images have a separate lifecycle from vCenter and the Supervisor nodes.

What is a STIG Readiness Guide?

More information about STIG Readiness Guides can be found at:
<https://www.vmware.com/docs/vmw-stig-program-overview>

What does the “status” column in the control list tables mean?

Status Definitions	
Passed	The compliance check passed.
Failed	The compliance check failed.
Not Applicable	The control was determined to be N/A in this context.
Not Reviewed	These controls were skipped as the conditions of the test did not exist on the system or require manual review and count as failures unless otherwise attested to manually.

Appendix: Full Control List – Photon OS 5.0

Control ID	Title	Status
PHTN-50-000003	The Photon operating system must audit all account creations.	Passed
PHTN-50-000004	The Photon operating system must enforce the limit of three consecutive invalid logon attempts by a user during a 15-minute time period.	Passed
PHTN-50-000005	The Photon operating system must display the Standard Mandatory DOD Notice and Consent Banner before granting local or remote access to the system.	Not Applicable
PHTN-50-000007	The Photon operating system must limit the number of concurrent sessions to ten for all accounts and/or account types.	Passed
PHTN-50-000012	The Photon operating system must monitor remote access logins.	Passed
PHTN-50-000013	The Photon operating system must have the OpenSSL FIPS provider installed to protect the confidentiality of remote access sessions.	Passed
PHTN-50-000014	The Photon operating system must configure auditd to log to disk.	Passed
PHTN-50-000016	The Photon operating system must enable the auditd service.	Passed
PHTN-50-000019	The Photon operating system must be configured to audit the execution of privileged functions.	Passed
PHTN-50-000021	The Photon operating system must alert the ISSO and SA in the event of an audit processing failure.	Passed
PHTN-50-000026	The Photon operating system must protect audit logs from unauthorized access.	Passed
PHTN-50-000030	The Photon operating system must allow only authorized users to configure the auditd service.	Passed
PHTN-50-000031	The Photon operating system must generate audit records when successful/unsuccessful attempts to access privileges occur.	Passed
PHTN-50-000035	The Photon operating system must enforce password complexity by requiring that at least one uppercase character be used.	Passed
PHTN-50-000036	The Photon operating system must enforce password complexity by requiring that at least one lowercase character be used.	Passed
PHTN-50-000037	The Photon operating system must enforce password complexity by requiring that at least one numeric character be used.	Passed
PHTN-50-000038	The Photon operating system must require the change of at least eight characters when passwords are changed.	Passed
PHTN-50-000039	The operating system must store only encrypted representations of passwords.	Passed
PHTN-50-000040	The Photon operating system must not have the telnet package installed.	Passed

Control ID	Title	Status
PHTN-50-000041	The Photon operating system must enforce one day as the minimum password lifetime.	Passed
PHTN-50-000042	The Photon operating systems must enforce a 90-day maximum password lifetime restriction.	Passed
PHTN-50-000044	The Photon operating system must enforce a minimum 15-character password length.	Passed
PHTN-50-000046	The Photon operating system must require authentication upon booting into single-user and maintenance modes.	Passed
PHTN-50-000047	The Photon operating system must disable unnecessary kernel modules.	Waivered
PHTN-50-000049	The Photon operating system must not have duplicate User IDs (UIDs).	Passed
PHTN-50-000059	The Photon operating system must use mechanisms meeting the requirements of applicable federal laws, Executive orders, directives, policies, regulations, standards, and guidance for authentication to a cryptographic module.	Passed
PHTN-50-000067	The Photon operating system must restrict access to the kernel message buffer.	Passed
PHTN-50-000068	The Photon operating system must be configured to use TCP syncookies.	Passed
PHTN-50-000069	The Photon operating system must terminate idle Secure Shell (SSH) sessions after 15 minutes.	Not Applicable
PHTN-50-000073	The Photon operating system /var/log directory must be restricted.	Passed
PHTN-50-000074	The Photon operating system must reveal error messages only to authorized users.	Passed
PHTN-50-000076	The Photon operating system must audit all account modifications.	Passed
PHTN-50-000078	The Photon operating system must audit all account removal actions.	Passed
PHTN-50-000079	The Photon operating system must implement only approved ciphers to protect the integrity of remote access sessions.	Passed
PHTN-50-000080	The Photon operating system must initiate session audits at system start-up.	Passed
PHTN-50-000082	The Photon operating system must protect audit tools from unauthorized access.	Passed
PHTN-50-000085	The Photon operating system must limit privileges to change software resident within software libraries.	Passed
PHTN-50-000086	The Photon operating system must enforce password complexity by requiring that at least one special character be used.	Passed
PHTN-50-000092	The Photon operating system must use cryptographic mechanisms to protect the integrity of audit tools.	Passed

Control ID	Title	Status
PHTN-50-000093	The operating system must automatically terminate a user session after inactivity time-outs have expired.	Passed
PHTN-50-000105	The Photon operating system must enable symlink access control protection in the kernel.	Passed
PHTN-50-000107	The Photon operating system must audit the execution of privileged functions.	Passed
PHTN-50-000108	The Photon operating system must automatically lock an account until the locked account is released by an administrator when three unsuccessful logon attempts in 15 minutes occur.	Passed
PHTN-50-000110	The Photon operating system must allocate audit record storage capacity to store audit records when audit records are not immediately sent to a central audit record storage facility.	Passed
PHTN-50-000112	The Photon operating system must immediately notify the SA and ISSO when allocated audit record storage volume reaches 75% of the repository maximum audit record storage capacity.	Passed
PHTN-50-000127	The Photon operating system must install AIDE to detect changes to baseline configurations.	Passed
PHTN-50-000130	The Photon operating system TDNF package management tool must cryptographically verify the authenticity of all software packages during installation.	Passed
PHTN-50-000133	The Photon operating system must require users to reauthenticate for privilege escalation.	Passed
PHTN-50-000160	The Photon operating system must implement address space layout randomization to protect its memory from unauthorized code execution.	Passed
PHTN-50-000161	The Photon operating system must remove all software components after updated versions have been installed.	Passed
PHTN-50-000173	The Photon operating system must generate audit records when successful/unsuccessful logon attempts occur.	Passed
PHTN-50-000175	The Photon operating system must be configured to audit the loading and unloading of dynamic kernel modules.	Passed
PHTN-50-000182	The Photon operating system must implement NIST FIPS-validated cryptography for the following: to provision digital signatures, to generate cryptographic hashes, and to protect unclassified information requiring confidentiality and cryptographic protection in accordance with applicable federal laws, Executive Orders, directives, policies, regulations, and standards.	Passed
PHTN-50-000184	The Photon operating system must prevent the use of dictionary words for passwords.	Passed
PHTN-50-000185	The Photon operating system must enforce a delay of at least four seconds between logon prompts following a failed logon attempt in login.defs.	Passed

Control ID	Title	Status
PHTN-50-000186	The Photon operating system must ensure audit events are flushed to disk at proper intervals.	Passed
PHTN-50-000187	The Photon operating system must define default permissions for all authenticated users in such a way that the user can only read and modify their own files.	Passed
PHTN-50-000188	The Photon operating system must configure Secure Shell (SSH) to disallow HostbasedAuthentication.	Passed
PHTN-50-000192	The Photon operating system must be configured to use the pam_faillock.so module.	Not Applicable
PHTN-50-000193	The Photon operating system must prevent leaking information of the existence of a user account.	Passed
PHTN-50-000194	The Photon operating system must audit logon attempts for unknown users.	Passed
PHTN-50-000195	The Photon operating system must include root when automatically locking an account until the locked account is released by an administrator when three unsuccessful logon attempts occur during a 15-minute time period.	Passed
PHTN-50-000196	The Photon operating system must persist lockouts between system reboots.	Passed
PHTN-50-000197	The Photon operating system must be configured to use the pam_pwquality.so module.	Passed
PHTN-50-000199	The Photon operating system TDNF package management tool must cryptographically verify the authenticity of all software packages during installation for all repos.	Passed
PHTN-50-000200	The Photon operating system must configure the Secure Shell (SSH) SyslogFacility.	Passed
PHTN-50-000201	The Photon operating system must enable Secure Shell (SSH) authentication logging.	Passed
PHTN-50-000203	The Photon operating system must terminate idle Secure Shell (SSH) sessions.	Not Applicable
PHTN-50-000204	The Photon operating system must audit all account modifications.	Passed
PHTN-50-000206	The Photon operating system must enforce a delay of at least four seconds between logon prompts following a failed logon attempt.	Passed
PHTN-50-000207	The Photon operating system must configure Secure Shell (SSH) to disallow authentication with an empty password.	Passed
PHTN-50-000208	The Photon operating system must configure Secure Shell (SSH) to disable user environment processing.	Passed
PHTN-50-000209	The Photon operating system must create a home directory for all new local interactive user accounts.	Passed

Control ID	Title	Status
PHTN-50-000210	The Photon operating system must disable the debug-shell service.	Passed
PHTN-50-000211	The Photon operating system must configure Secure Shell (SSH) to disallow Generic Security Service Application Program Interface (GSSAPI) authentication.	Passed
PHTN-50-000212	The Photon operating system must configure Secure Shell (SSH) to disable X11 forwarding.	Passed
PHTN-50-000213	The Photon operating system must configure Secure Shell (SSH) to perform strict mode checking of home directory configuration files.	Passed
PHTN-50-000214	The Photon operating system must configure Secure Shell (SSH) to disallow Kerberos authentication.	Passed
PHTN-50-000215	The Photon operating system must configure Secure Shell (SSH) to disallow compression of the encrypted session stream.	Passed
PHTN-50-000216	The Photon operating system must configure Secure Shell (SSH) to display the last login immediately after authentication.	Passed
PHTN-50-000217	The Photon operating system must configure Secure Shell (SSH) to ignore user-specific trusted hosts lists.	Passed
PHTN-50-000218	The Photon operating system must configure Secure Shell (SSH) to ignore user-specific known_host files.	Not Applicable
PHTN-50-000219	The Photon operating system must configure Secure Shell (SSH) to limit the number of allowed login attempts per connection.	Passed
PHTN-50-000220	The Photon operating system must configure Secure Shell (SSH) to restrict AllowTcpForwarding.	Passed
PHTN-50-000221	The Photon operating system must configure Secure Shell (SSH) to restrict LoginGraceTime.	Not Applicable
PHTN-50-000222	The Photon operating system must be configured so that the x86 Ctrl-Alt-Delete key sequence is disabled on the command line.	Passed
PHTN-50-000223	The Photon operating system must not forward IPv4 or IPv6 source-routed packets.	Passed
PHTN-50-000224	The Photon operating system must not respond to IPv4 Internet Control Message Protocol (ICMP) echoes sent to a broadcast address.	Passed
PHTN-50-000225	The Photon operating system must prevent IPv4 Internet Control Message Protocol (ICMP) redirect messages from being accepted.	Passed
PHTN-50-000226	The Photon operating system must prevent IPv4 Internet Control Message Protocol (ICMP) secure redirect messages from being accepted.	Passed
PHTN-50-000227	The Photon operating system must not send IPv4 Internet Control Message Protocol (ICMP) redirects.	Passed
PHTN-50-000228	The Photon operating system must log IPv4 packets with impossible addresses.	Passed

Control ID	Title	Status
PHTN-50-000229	The Photon operating system must use a reverse-path filter for IPv4 network traffic.	Passed
PHTN-50-000231	The Photon operating system must not perform IPv4 packet forwarding.	Waivered
PHTN-50-000232	The Photon operating system must send TCP timestamps.	Passed
PHTN-50-000233	The Photon operating system must be configured to protect the Secure Shell (SSH) public host key from unauthorized modification.	Passed
PHTN-50-000234	The Photon operating system must be configured to protect the Secure Shell (SSH) private host key from unauthorized access.	Passed
PHTN-50-000235	The Photon operating system must enforce password complexity on the root account.	Passed
PHTN-50-000236	The Photon operating system must disable systemd fallback DNS.	Passed
PHTN-50-000237	The Photon operating system must configure AIDE to detect changes to baseline configurations.	Passed
PHTN-50-000239	The Photon operating system must implement only approved Message Authentication Codes (MACs) to protect the integrity of remote access sessions.	Passed
PHTN-50-000241	The Photon operating system must install rsyslog for offloading of audit logs.	Passed
PHTN-50-000242	The Photon operating system must enable the rsyslog service.	Passed
PHTN-50-000244	The Photon operating system must enable hardlink access control protection in the kernel.	Passed
PHTN-50-000245	The Photon operating system must mount /tmp securely.	Waivered
PHTN-50-000246	The Photon operating system must restrict core dumps.	Passed
PHTN-50-000247	The Photon operating system must not allow empty passwords.	Passed
PHTN-50-000266	The Photon operating system must generate audit records for all account creations, modifications, disabling, and termination events that affect /etc/security/opasswd.	Waivered
PHTN-50-000267	The Photon operating system must be configured to use the pam_deny.so module.	Not Applicable

